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MAT 213: Differential Equation

First Order Ordinary Differential Equations: Exact Differential Equations

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Given a differential equation

$$M(x,y)dx + N(x,y)dy = 0, \qquad 1$$

if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

then the differential equation is exact.

Example. Show whether
$(6xy^2 + 4x^3y)dx + (6x^2y + x^4 + e^y)dy = 0$ is exact or not.

Solution $M(x,y) = 6xy^2 + 4x^3y$, $N(x,y) = 6x^2y + x^4 + e^y$

$$\frac{\partial M}{\partial y} = 12xy + 4x^3, \ \frac{\partial N}{\partial x} = 12xy + 4x^3 \quad \therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\Rightarrow$ Equation is exact.

Problems

Determine if the following differential equations are exact or not

1. $2tyy' + 2t + y^2 = 0$
2. $\sin t\, y' + t^2 e^y y' - y' = -y\cos t - 2te^y$
3. $(y\cos x + 2xe^y) + (\sin x + x^2 e^y - 1)y' = 0$
4. $y^3 + 3xy^2 y' = 0$

Steps for Solving an Exact Equation

1. Determine if the ODE is exact
2. Split the ODE into two equations

$$\frac{\partial F}{\partial x} = M(x, y) \tag{2}$$

$$\frac{\partial F}{\partial y} = N(x, y) \tag{3}$$

3. we first integrate $M(x, y)$ with respect to x and write

$$F(x, y) = \int M(x, y)dx + g(y), \tag{4}$$

were $g(y)$ is an arbitrary constant.

4 Differentiate $F(x, y)$ with respect to y

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}\int M(x, y)dx + g'(y) \qquad 5$$

5 Compare equation (5) and (3), so that

$$g'(y) = N(x, y) - \frac{\partial}{\partial y}\int M(x, y)dx \qquad 6$$

6 Integrate equation (6) with respect to y

$$g(y) = \int \left(N(x, y) - \frac{\partial}{\partial y}\int M(x, y)dx \right) dy \qquad 7$$

7 Substitute $g(y)$ in equation (4) to obtain the general solution

$$F(x,y) = \int M(x,y)dx + \int \left(N(x,y) - \frac{\partial}{\partial y} \int M(x,y)dx \right) dy = C$$

Example. Solve the exact ODE
$(6xy^2 + 4x^3y)dx + (6x^2y + x^4 + e^y)dy = 0.$

Solution $M(x,y) = 6xy^2 + 4x^3y$, $N(x,y) = 6x^2y + x^4 + e^y$

$$F(x,y) = \int (6xy^2 + 4x^3y)dx = 3x^2y^2 + x^4y + g(y)$$

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}(3x^2y^2 + x^4y + g(y)) = 6x^2y + x^4 + g'(y)$$

Comparing to $N(x,y)$, $g'(y) = e^y \quad \Rightarrow \quad g(y) = e^y$

$F(x, y) = 3x^2y^2 + x^4y + e^y$ and the general solution is $3x^2y^2 + x^4y + e^y = c$

Problems

Solve the following equations:

1. $2x(3x + y - ye^{-x^2})dx + (x^2 + 3y^2 + e^{-x^2})dy = 0$
2. $(y\cos x + 2xe^y) + (\sin x + x^2e^y - 1)y' = 0$
3. $2xy - 9x^2 + (2y + x^3 + 1)y' = 0, \quad y(0) = -3$
4. $2xy^2 + 4 = 2(3 - x^2y)y' \quad y(-1) = 8$
5. Show that any separable equation

$$M(x) + N(y)y' = 0$$

is also exact.

Integrating Factors

Given the differential equation

$$M(x,y)dx + N(x,y)dy = 0. \tag{8}$$

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, the differential equation is not exact.

The differential equation can be made exact by multiplying it by an integrating factor i.e.

$$\mu(x,y)M(x,y)dx + \mu(x,y)N(x,y)dy = 0, \tag{9}$$

is exact.

To find an integrating factor, use the exactness criteria

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x},$$

expanding

$$M\frac{\partial \mu}{\partial y} + \mu\frac{\partial M}{\partial y} = N\frac{\partial \mu}{\partial x} + \mu\frac{\partial N}{\partial x}$$

$$\Rightarrow \mu\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y} \qquad 10$$

(10) is a PDE which is difficult to solve. The integrating factor can only be found for special cases only when μ is a function of only one variables x or y, instead of both.

if $\frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)$ is a function of x only, then

$$\mu(x)=exp\left[\int \frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)dx\right]. \tag{11}$$

If $\frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)$ is a function of y only, then

$$\mu(y)=exp\left[\int \frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)dy\right]. \tag{12}$$

Example. Solve $xydx + (2x^2 + 3y^2 - 20)dy = 0$.

Solution. $\frac{\partial M}{\partial y} = x, \quad \frac{\partial N}{\partial x} = 4x, \quad \therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$ depends on x and y.

$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = \frac{4x - x}{xy} = \frac{3}{y}$ depends on y only.

$\mu(y) = e^{\int \frac{3}{y} dy} = e^{3lny} = y^3$ Multiplying the differential equation by the integrating factor gives

$$xy^4 dx + (2x^2y^3 + 3y^5 - 20y^3)dy = 0$$

$$\frac{\partial M}{\partial y} = 4xy^3, \quad \frac{\partial N}{\partial x} = 4xy^3, \quad \therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$F(x,y) = \int xy^4 dx = \frac{x^2y^4}{2} + g(h)$$

$\frac{\partial F}{\partial y} = 2x^2y^3 + g'(y)$. Comparing this to $N(x,y)$,

$g'(y) = 3y^5 - 20y^3$ and $g(y) = \int 3y^5 - 20y^3 dy = \frac{y^6}{2} - 5y^4$

$F(x,y) = \frac{x^2y^4}{2} + \frac{y^6}{2} - 5y^4$ and the general solution is $\frac{x^2y^4}{2} + \frac{y^6}{2} - 5y^4 = c$.

Problems Solve the following

1. $(3xy + y^2) + (x^2 + xy)y' = 0$
2. $xdx + (x^2y + 4y)dy = 0, \quad y(4) = 0.$
3. $(x^2 + y^2 - 5)dx = (y + xy)dy, \quad y(0) = 1.$

Find the value of b for which the following equations are exact and then solve then using the value of b

4. $(xy^2 + bx^2y) + (x + y)x^2y' = 0.$
5. $(ye^{2xy} + x) + bxe^{2xy}y' = 0.$